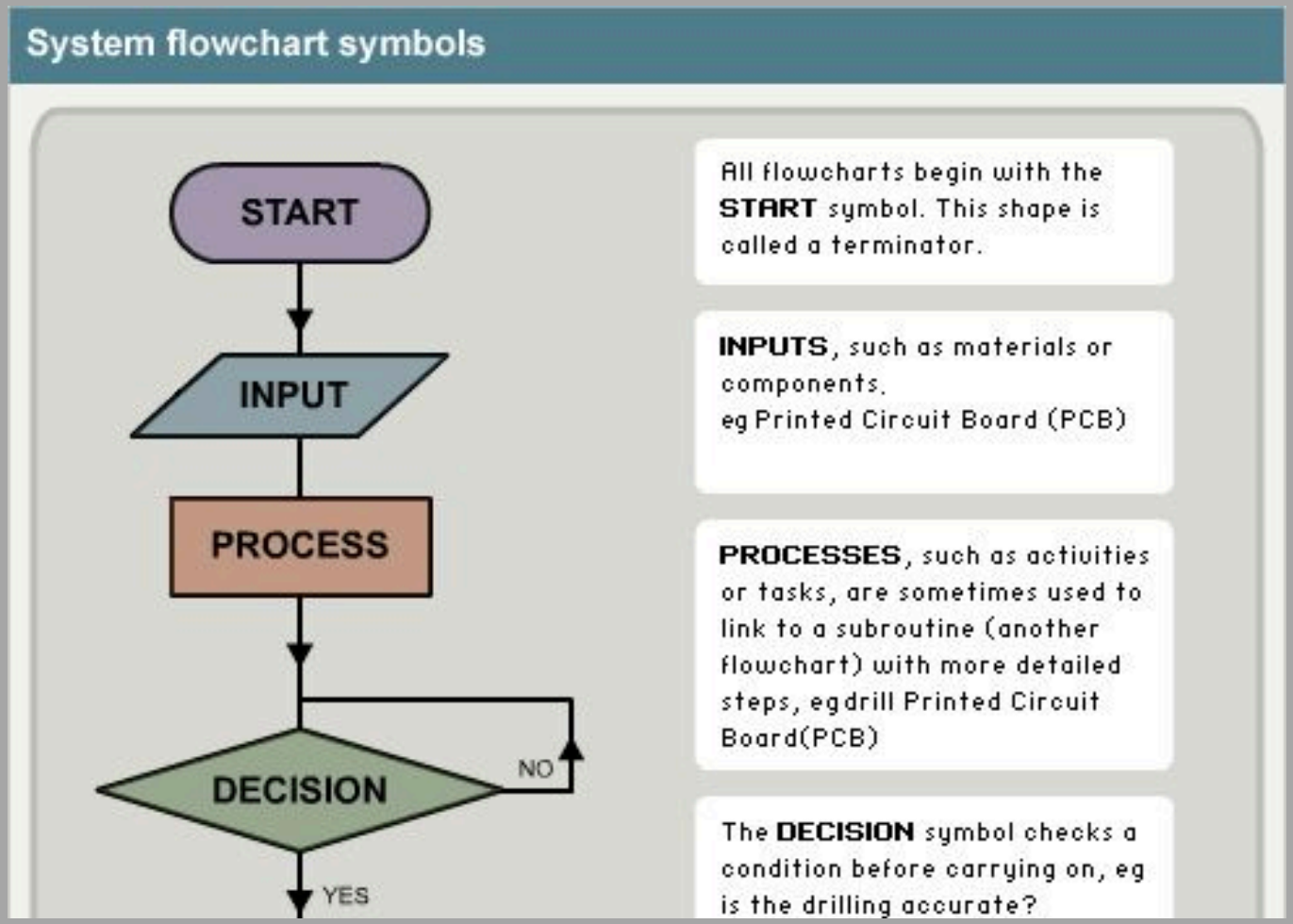
foundation of computing

Flow chart

A flowchart is a visual diagram that uses standardized symbols and arrows to represent a process, workflow, or algorithm. It breaks down complex sequences into a series of logical steps, making it easier to understand, document, and improve processes. Common symbols include ovals for start/end, rectangles for processes, and diamonds for decisions.

flow chart symbols and meaning



Algorithm

An algorithm is a step-by-step set of instructions or rules to solve a problem or perform a task, converting an input into a desired output. Like a recipe, it provides a precise procedure that can be followed by a human or, more commonly, a computer to achieve a consistent and reliable result. Algorithms are fundamental to computer science and are used in everything from simple calculations to complex operations like traffic navigation, personalized recommendations, and data processing.

How Algorithms work

- **Input:**
- Algorithms begin with one or more specific inputs, which is the data to be processed.
- **Processing:**
- A series of well-defined, logical steps are performed on the input data. These steps can involve simple operations, decision-making (branching), or repetition (loops).
- **Output:**
- The algorithm produces a result or a solution based on the input and the executed steps.

Key characteristics

- **Definiteness:** Each step is clear and unambiguous, leaving no room for interpretation.
- **Finiteness:** An algorithm will always finish after a limited number of steps.
- **Effectiveness:** Each instruction must be basic enough to be carried out in practice.
- **Input/Output:** There are zero or more well-defined inputs and one or more well-defined outputs.

Real -world Example

- **Navigation apps:**
- Use algorithms to find the fastest route between two points, factoring in traffic and road closures.
- **Social media:**
- Employ algorithms to curate your feed, showing you content based on your past activity and interests.
- **Search engines:**
- Use proprietary algorithms to rank websites and provide relevant results for your search queries.

Algorithm Example

Step 1: Start

Step 2: Create a variable to receive the user's email address

Step 3: Clear the variable in case it's not empty

Step 4: Ask the user for an email address

Step 5: Store the response in the variable

Step 6: Check the stored response to see if it is a valid email address

Step 7: Not valid? Go back to Step 3.

Example program

```
Enter the number of rows of First Matrix: 2
Enter the number of columns of First Matrix: 2
Enter the elements of First Matrix:
A[0][0]: 2
A[0][1]: 3
A[1][0]: 4
A[1][1]: 5
Enter the number of rows of Second Matrix: 2
Enter the number of columns of Second Matrix: 2
Enter the elements of Second Matrix:
B[0][0]: 5
B[0][1]: 2
B[1][0]: 3
B[1][1]: 1
First Matrix:
[[2. 3.]
 [4. 5.]]
Second Matrix:
[[5. 2.]
 [3. 1.]]
Resultant Matrix:
[[19.  7.]
```

Output for the program

- First Matrix:
- [2. 3.]
- [4. 5.]

- Second Matrix:
- [5. 2.]
- [3. 1.]

- Resultant Matrix:
- [19. 7.]
- [35. 13.]

THANK
YOU